Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

School of Mathematics

M.Sc Mathematics and Computing (First Year) Course Code: PMA 208

Course Name: Numerical Analysis

May 13, 2023

Time: 3 Hours, M. Marks: 40 Name of Faculty: Amit Kumar

Note: Attempt all the questions.

Q1. Perform one iteration of Newton Raphson method, find the complex root of the equation $1 + (x + iy)^2 = 0$ by taking $x_o = y_0 = 0.5$ (4)

Q2. Using Power method, find the largest eigen value up to two decimal places of the matrix $A = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$. (4)

Q3. Reduce the symmetric matrix $A = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to tri-diagonal form by Householder's method. (4)

Q4. Using the expression, $P_{(2n+1)}(x) = \sum_{i=0}^{n} \left((-2l_i'(x_i)x + 1 + 2x_i l_i'(x_i))(l_i(x))^2 \right) f(x_i) + \sum_{i=0}^{n} \left((x - x_i)(l_i(x))^2 \right) f'(x_i)$. Determine a piecewise cubic Hermite interpolating polynomial by considering $f(0) = 1, f'(0) = 0, f(1) = 3, f'(1) = 6, f(2) = 21, f'(2) = 36$. (4)

Q5. Compute the value of integral $I = \int_1^2 e^{-x^2} dx$ with the help of Simpson's one-third rule with $n = 2, 4, 8$. Then, use Romberg integration to improve these values. (4)

Q6. Compute the value of integral $I = \int_0^1 \int_1^2 \frac{2y}{(1+x^2)} dxdy$ with the help of Trapezoidal rule by considering four equal subintervals for the variable x and four equal subintervals for the variable y. (4)

Q7. Derive Runge-Kutta second order formula to solve the equation $\frac{dy}{dx} = f(x, y), y(x_0) = y_o$. (4)

Q8. Using finite difference method solve the differential equation $y'' - xy' + 2y = x + 4$, $0 \le x \le 0.6$ subject to the boundary conditions $y'(0) = 1$ and $y(0.6) = 1.96$ by taking three equal subintervals or step size $h = 0.20$. Use Central difference approximations. (4)

Q9. Derive Milne-Simpson method to find $y(x)$ for the equation $\frac{dy}{dx} = f(x, y)$ by considering $(n + 1)$ points $x_i; i = 0,1,2, \ldots, n$. (4)

Q10. Solve the differential equation with the help of Shooting method $y'' - y = x, 0 \le x \le 0.2, y(0) = 0, y(0.1) = -0.2$. Solve the corresponding initial value problems by Runge-Kutta fourth order method with step size $h = 0.1$. (4)